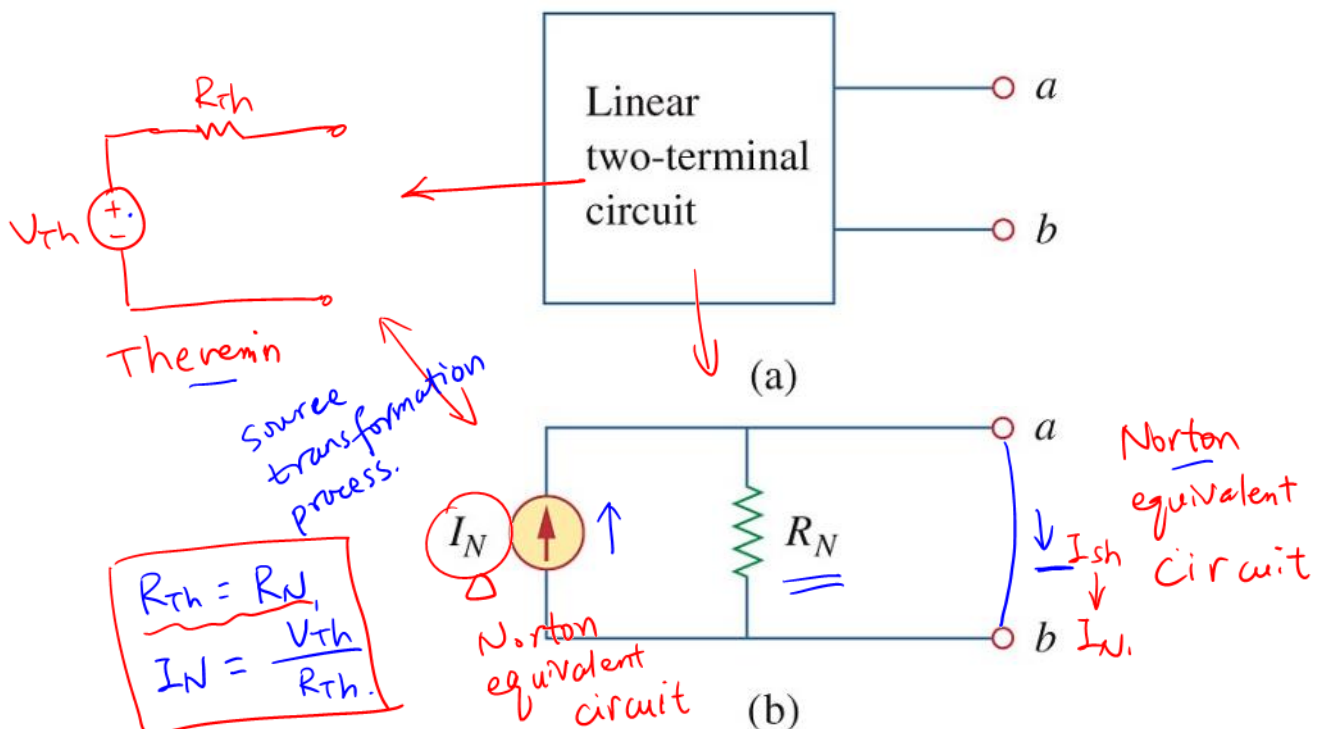


L#19: Norton's Theorem



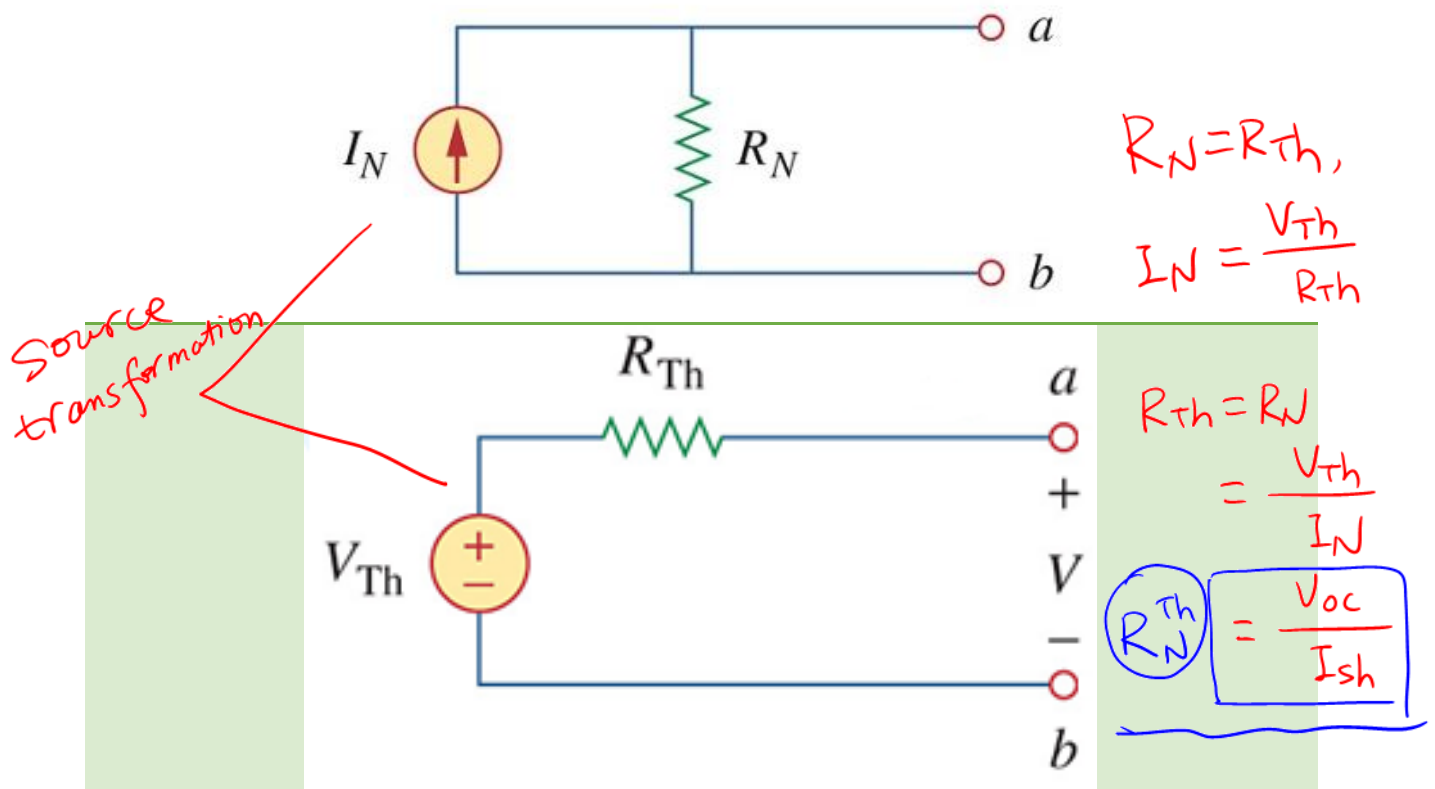
I_N : Short circuit current wrt the terminals a and b . the current flows from a to b .

R_N : $R_N = R_{th}$.

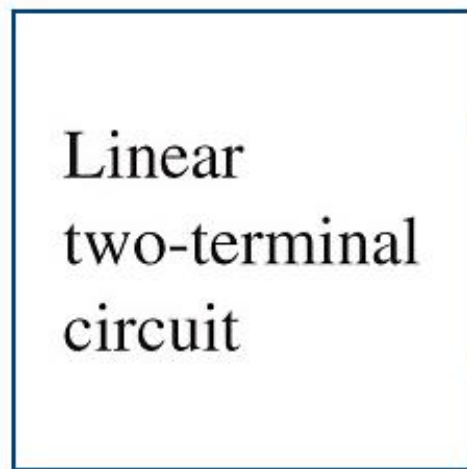
how to obtain I_N , R_N .

- by measurement at the interested terminals. $a.b$
- by analysis/calculation.

Norton's Theorem and Thévenin's Theorem



Determine I_N



Determine R_N

$= R_{th}$,
only with independent sources.

Method #1:

- ① switch off all ind. sources.
- ② analyse resistive network for R_N .

Method #2:

- circuit includes dependent source.
- ① switch off all ind. sources,
 - ② add test current/voltage.

Method #3:

③ $R_N = R_{th} = \frac{V_T}{I_T}$.

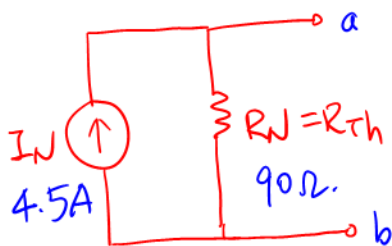
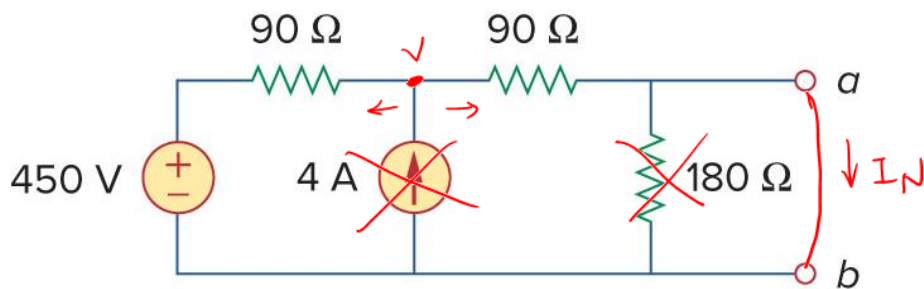
- $R_N = R_{th} = V_{oc} / i_{sc}$

- Works for circuits with or without dependent sources. powerful!

Example 11: Norton

- Obtain the Norton equivalent circuit to the left of terminals $a-b$.

With no dependent source.



① switch off ind. sources.

$$R_N = 180 \parallel (90 + 90) = 90 \Omega.$$

Try superposition Theorem.

(1) contribution from 4A, let $V=0$.

$$I_N^{4A} = \underline{2A}.$$

(2) contribution from 450V, let $I=0$.

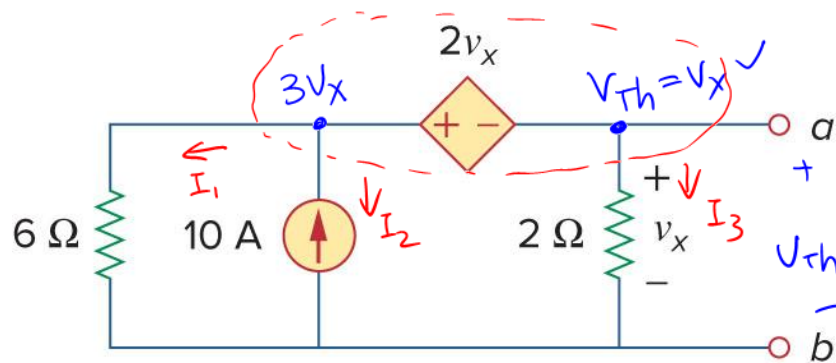
$$I_N^{450V} = \frac{450}{(90+90)} = \frac{5}{2} = \underline{2.5A}.$$

$$I_N = 4.5A$$

Example 12: Norton

- Obtain the Norton equivalent circuit with respect to terminals $a-b$.

I_N , R_N .



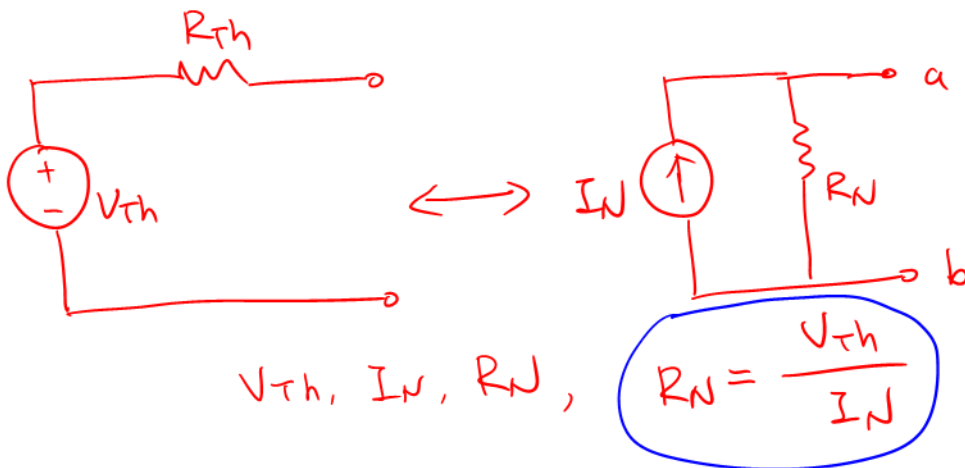
$I_N = 10A$

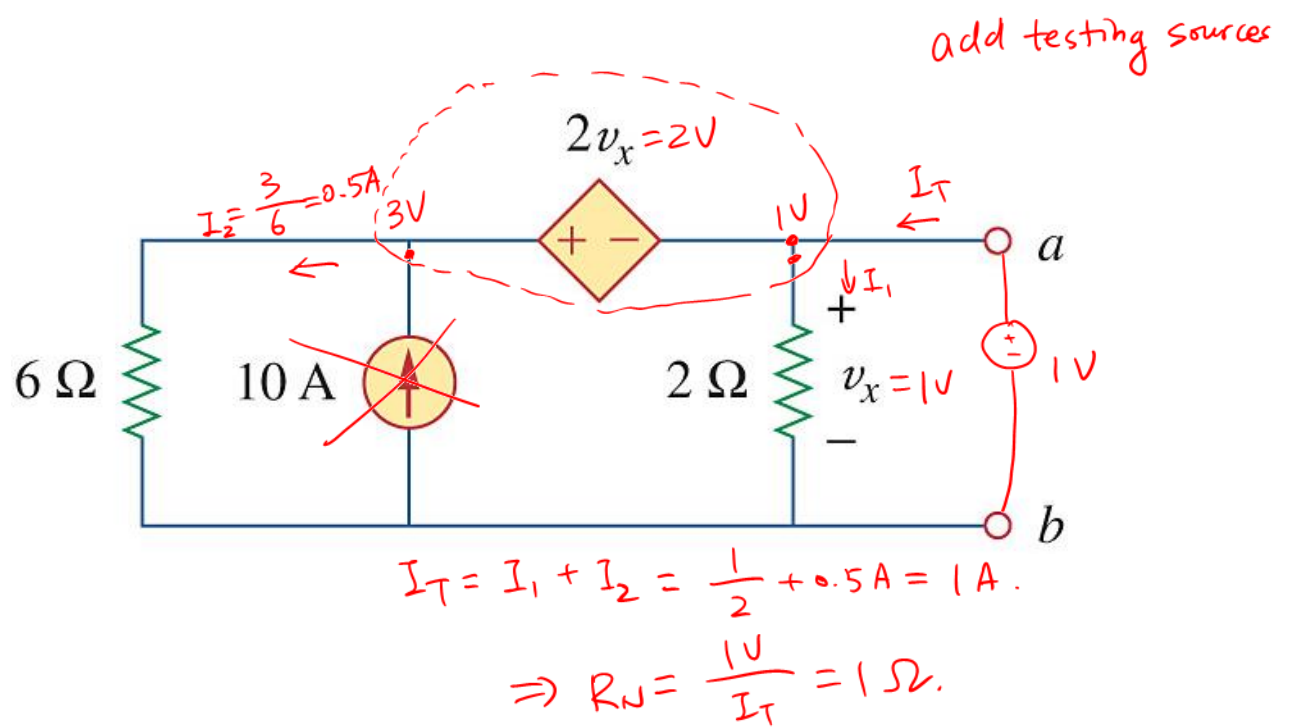
$$I_N = 10A, \quad \underline{R_N} = \frac{V_{oc} = V_{Th}}{I_{sh} = I_N}$$

$$I_1 + I_2 + I_3 = 0$$

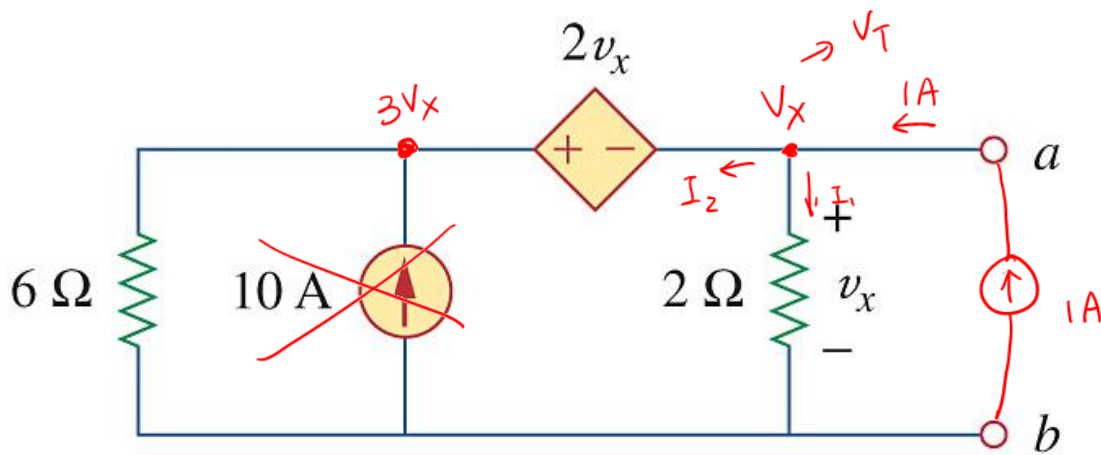
$$\frac{3V_x}{6} - 10 + \frac{V_x}{2} = 0 \Rightarrow V_x = 10V.$$

$$\Rightarrow R_N = \frac{V_{Th}}{I_N} = 1\Omega.$$





add current source.



$$I_1 + I_2 = 1A$$

$$\frac{V_x}{2} + \frac{V_x + 2V_x}{6} = 1$$

$$\Rightarrow V_x = 1V = V_T.$$

$$\Rightarrow R_N = \frac{1V}{1A} = 1\Omega.$$

Example 13: Norton

Determine the Norton equivalent circuit. wrt terminals a-b.

